

The secret of voice: How acoustic characteristics affect video creators' performance on Bilibili

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ABSTRACT

The importance of voice has been well acknowledged in sensory decision-making. Yet, past literature on video creators' performance did not shed much light on the impact of video creators' acoustic characteristics. Building on signaling theory of portfolios, we examine how the acoustic characteristics of a video creator and the signals of video quantity affect the number of likes a video creator receives and the change in the number of the creator's followers. Using a longitudinal dataset obtained from Bilibili through automated speech recognition analytics and text analysis approach, econometrics models are employed to test the research model. Findings indicate that loudness variability of a video creator significantly increases the number of likes the creator receives, and exerts a positive effect on the change in the number of the creator's followers. In contrast, vocal pitch has a significant positive impact on the number of likes but a significant negative impact on the change in the number of followers. Results further suggest that a video creator's performance varies according to the signals of video quantity, i.e., the number of weekly published videos and the average length of videos. Our results could offer theoretical contributions and practical insights for video creators to enhance their performance by adjusting the acoustic characteristics and the signals of video quantity wisely.

1. Introduction

The continuous proliferation of web 2.0 has accelerated the prevalence of video sharing platforms to afford the sharing of user-generated videos [1,2]. As more complex and hybrid, online user-generated videos convey valuable information through visual stimuli and acoustic arousals [3,4]. According to a recent report, online video's global market size is expected to be valued at over \$149.34 billion by 2026.¹ Many video sharing platforms have emerged in recent years. For example, video creators can consistently post videos on YouTube, delivering 1 billion hours of videos per day [5]. Moreover, statistics show that the number of monthly active users on Bilibili, a user-generated video platform, keeps increasing and has reached 267 million in the third quarter of 2021.²

Given the vast number of videos generated on the platforms, not all

of them receive the desired attention from the audience, especially for those video creators with limited traffic-inducing channels [6]. Research has shown that loyal program users are more likely to be notified of the updates from focal streamers [7], and online traffic from subscribed viewers garners high conversions [8]. In addition, video creators with better online performance can earn great attention and create tremendous business value for endorsing enterprises [9]. Online performance metrics, such as the number of likes and followers, are critical to video creators' traffic and monetary gains from the platforms [10–12]. As a result, video creators are under fierce competition to make their videos go viral and get more likes [13,14], and need great efforts to convert viewers into followers [8,15].

During viewers' decision-making processes to click a “like” button or follow an account, video creators' characteristics, the quality of video content, and the interactivity between video creators and viewers are

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¹ <https://www.alliedmarketresearch.com/video-streaming-market>.

² <https://www.statista.com/statistics/1109108/bilibili-average-monthly-active-users/>.

the frequently documented influencing factors [15]. It is worth noting that voice can also impact the perceptions and behaviors of listeners toward the speakers [16,17]. Existing research has highlighted the importance of voice in influencing individuals' preferences and choices [18–21]. However, there is a lack of research exploring the impact of the acoustic characteristics of video creators.

In traditional mass media communication, it is imperative to provide broadcasters with proper training to exhibit appropriate appearances and voice expressions [22–24]. Similarly, serving as a means of communication, creators on video-sharing platforms should also be mindful of utilizing acoustic characteristics effectively. Different from traditional user-generated content with a textual nature, video creators may have unique acoustic characteristics in user-generated videos. Thus, it is gaining greater importance for video creators to modulate the acoustic characteristics of videos catering to the viewers' preferences [25]. Needless to say, user-generated video creators are usually amateurs who lack professional training to behave in suitable acoustic characteristics [2] and require related knowledge. Therefore, this paper examines two acoustic characteristics of video creators, namely loudness variability and vocal pitch, which may significantly affect viewers' perceptions and behaviors but have been neglected in the past promotion of creators' performance [26]. Here, loudness variability refers to the variations of sound amplitude, as that speakers often express their emotions by fluctuating their voices [27]. Vocal pitch is defined as the frequency band where most energy is concentrated [28].

According to recent explorations on signal portfolios [29,30], acoustic characteristics of video creators work together with the signals of video quantity to influence viewers' decisions. The effect of acoustics characteristics on video creators' online performance may be different varying with the signals of video quantity [31], such as the number of weekly published videos and the average length of videos. Thus, this paper examines the boundary conditions of the associations between the acoustic characteristics of video creators and viewers' liking and following decisions. Not confining to the isolated impact of acoustic characteristics [16,26], we aim to understand under what conditions and in what way the signals of video quantity interact with the acoustic characteristics of video creators. The following research questions are addressed:

Research Question 1. How do a video creator's acoustic characteristics (i.e., loudness variability and vocal pitch) affect the number of likes the creator receives on the video sharing platforms?

Research Question 2. How do a video creator's acoustic characteristics (i.e., loudness variability and vocal pitch) affect the change in the number of followers³ the creator receives on the video sharing platforms?

Research Question 3. How do the signals of video quantity moderate the impact of acoustic characteristics on (a) the number of likes and (b) the change in the number of followers the creator receives?

To answer these questions, a longitudinal dataset was obtained from a popular user-generated video sharing platform in China, Bilibili. Speech recognition analytics were used to extract the acoustic characteristics of videos, and text analysis was adopted for the measurement of video quality and video topics. Econometrics models are employed to test the research models. Several important findings were obtained. First, loudness variability has both a significantly positive impact on the number of likes and the number of follower changes, while vocal pitch has a significantly positive effect on the number of likes but a significantly negative impact on the number of follower changes. Second, the effects of loudness variability and vocal pitch vary according to the number of weekly published videos and the average length of videos. In

particular, the number of weekly published videos tends to strengthen the effects of loudness variability but weaken the negative effect of vocal pitch; while the average length of videos tends to strengthen the effects of loudness variability but weaken the positive effect of vocal pitch.

This work contributes to the previous literature in three aspects. First, although scholars have recognized the value of quality content and interactivity in improving online performance on video-sharing platforms [15,32,33], acoustic characteristics of videos received scant attention. Using the speech recognition analytics approach, we investigate the impact of video creators' voices on their performance, i.e., the number of likes and the change in the number of followers. Our paper not only supplements the existing literature on video creators' online performance [18,19,34] but also responds to the call for research using unstructured voice data in marketing practices [35].

Second, this paper provides empirical evidence for the specific acoustic characteristics that may take effect in video creators' online performance. For instance, loudness variability of the video creator is found to be positively related with the number of likes a video creator receives and the change in the number of the creator's followers; while vocal pitch is found to be positively related with the number of likes but is negatively related with the number of follower changes. It further consolidates previous arguments that “liking” is relatively an indicator of short-term engagement in online platforms and “following” is relatively a long-term engagement [10,36,37]. Considering the differences in the number of likes and the number of follower changes [11,38], video creators could adjust their acoustic strategies according to different performance goals.

Last, our paper is one of the first attempts focusing on the synergistic effect of acoustic characteristics of a video creator and the signals of video quantity on the performance of video creators, bearing significant implications for stakeholders to enhance the popularity of videos by modulating acoustic characteristics and adjusting content creation strategies. It goes beyond content and acoustic strategy on video creators' performance [15,37] and extends the antecedents of video creators' performance to the signals of portfolios.

2. Related works

This section reviews three streams of literature including the metrics for performance on video sharing platforms, various acoustic characteristics of video creators and the signaling theory.

2.1. Performance metrics on video sharing platforms

User-generated video refers to the video generated by non-professionals, which has become a powerful marketing channel to impact viewers' decisions [25,39,40–42]. User-generated video creators, sometimes called social media influencers, are individuals who create videos on digital platforms [43]. Video creators must embrace their viewers for more positive feedback and better performance [43] since positive feedback and good performance mostly induce valuable incoming traffics and monetary rewards [44–46].

The number of likes and followers are frequently used performance metrics on video sharing platforms [43,38,47]. These metrics are different [10,48]. The number of likes signals creators' social endorsements being able to deliver quality content with interactive affordances [47,49,50]. Influential creators are willing to get more followers as it signals a creator's popularity and reputation [8]. The number of followers also serves as a status cue afforded by others [51]. Prior research considered the number of likes as a quality indicator of a focal post and the number of followers as a popularity indicator in the long run [10,39]. It has been shown that both the number of likes and followers are the antecedents of content purchases/sales conversions on social platforms [51]. In addition, other studies have demonstrated that engagement metrics determine the search rankings of videos [52]. Collectively, performance metrics are important to video creators as the

³ The changes in the number of followers, increase in the number of followers, and number of follower changes are interchangeable hereinafter.

monetization abilities and prioritized recommendations are all closely related to the value of these metrics [15,50].

Therefore, video creators should do their best to improve their online performance by increasing the number of likes and followers. Previous studies have examined the influencing factors of performance metrics from the perspective of relationship herd and social network structures [37,49], interactive tactics and flow experience provision in attracting viewers [43,53], individuals' motivation to view Danmaku videos [54], video content strategies [2], and creators' characteristics [41]. Although the acoustic characteristics of video creators are important [18,19,28], few research have concentrated on the impact of acoustics characteristics on creators' performance.

2.2. Acoustic characteristics of video creator

Unstructured data has been widely applied in decision-making practices [21,35,55–57], and acoustic characteristic is one of the representatives of unstructured data regarding user-generated videos [16,58]. The importance of acoustic factors has been examined in the area of advertising and branding management, where marketers and managers use voice parameters to intensify emotions and deliver symbolic meanings [18,19,28]. Among various acoustic characteristics, voice loudness, vocal pitch, and loudness variability are frequently explored [55]. Specifically, faster, low-pitched, and louder voices (a.k.a. higher vocal intensity) signal more confidence because they are perceived with higher credibility and are generally associated with masculinity and dominance [19,31,57]. Moreover, speech rate is also a widely explored voice parameter [26,56]. Previous researchers have focused on the impact of basic video properties on individuals' continuous viewing intention and reported the negative impact of speech rate [56].

Given that voice loudness and speech rate from video creators primarily focus on different aspects of vocal delivery but may not directly relate to emotional expression or engagement, we thus turn our attention to loudness variability (the variations of sound amplitude) and vocal pitch (the frequency band where most of the energy is concentrated) [26,59]. These two factors have been extensively studied in relation to emotional expression, communication effectiveness, and audience engagement [16,28]. Therefore, they are particularly relevant for understanding how the acoustic features affect the number of likes and followers a video receives.

2.3. Signaling theory: from the perspective of signal portfolios

Signaling theory has been consistently used to explain situations where information asymmetries between two parties exist [60]. In such cases, signals can be applied to convey quality cues and network cues to potential decision-makers [50,61]. For example, product cues influence sale performance when customers are uncertainty about the quality of products [61].

Recent studies on signaling theory have proposed the concept of signal portfolios that combines different types of signals [29,51]. When information producers send multiple signals simultaneously, information receivers may experience a high-noise environment for decision-making [30]. Different types of signals in signal portfolios may substitute or complement each other in influencing information receivers' perceptions and behaviors [30]. In the online consulting market, it has been found that signals of seller and signals of market substitute each other to affect consultants' income positively [57]. Yet other studies posited that the synergic effects of signal types might vary with the credibility of signals [30]. Although some initial explorations exist on signal portfolios, ongoing studies are still required to extend the related work [30].

Our paper sets out to explore both the separate and joint effect of acoustic signals and quantity signals from video creators. *Video acoustic signals* convey non-verbal cues (i.e., loudness variability and vocal pitch)

to viewers in a procedural way. In other words, viewers need to spend time listening to the video thus to receive and process the acoustic cues. *Video quantity signals* are usually numerical information enable viewers to be aware of a video creator's efforts without long-standing deliberation (i.e., the number of weekly published videos, the average length of videos).

3. Research framework and hypotheses

Drawing on signaling theory [29,30], we propose that *acoustic signals* and *quantity signals* jointly impact the performance of video creators. Following the literature, two key online performance metrics, i.e., the number of likes and the change in the number of followers, are examined of particular focus [43,47,50]. We argue that the number of likes and the change in the number of followers are different in terms of their inner manifestations [10,39,62]. The number of likes is typically a quality indicator right after a focal video being watched [63], while the change in the number of followers is more like an accumulated popularity measure in the long term [51].

3.1. The effect of acoustic signals

Loudness variability can reflect emotional richness, embodied in the voice fluctuation of speaking [64]. Loudness variability entails up and down movement of voice loudness to maintain viewers' attention throughout. Videos with high loudness variability generally have specific words to be accentuated, certain sentences to be emphasized, thus eventually build an emotional connection between the viewers and the video content [65].

On user-generated video sharing platforms, liking behavior can be affectively triggered in the short term when viewers are immersed in the focal video content [38]. The number of likes reflects a creator's performance and displays viewer's short-term engagement [39,36]. Therefore, when the voice is emotionally rich (i.e., high loudness variability), viewers will be more engaged in the video content and vote for a like. However, synthetic voices may reduce viewers' perceived hedonic value during social interactions [66], as artificial intelligence-enabled voices generally have consistent loudness variability [67] and less positive emotional response [64]. Accordingly, we propose the following hypothesis:

H1. Loudness variability of a video creator is positively related to the number of likes he/she receives.

Vocal pitch refers to the fundamental frequency [26] of a user-generated video. A higher pitch tends to have a robust correlation to the harsh voice with a rasping sound [28], and is often seen in high-tension settings to immediately stimulate feelings of stress, anger, confidence, happiness, etc. [68]. For instance, gossip videos are generally in a high pitch which leads to immediate gratification, while science videos are mostly in a low pitch generating long-term knowledge benefits [12]. When the vocal pitch is high, the strong environmental stimuli could invoke individuals' emotional arousal and improve their temporary engagement in the focal activities [69,70]. Considering the short-term engagement orientation of liking behavior [39,71], we propose the following hypothesis:

H2. Vocal pitch of a video creator is positively related to the number of likes he/she receives.

The number of followers (i.e., subscribers) received by a video creator reflects his/her performance in the long term [10]. Viewers would expect long-lasting benefits when they follow a video creator [15]. However, such benefits will only be manifested if a user-generated video contains pertinent information with the right acoustic expressions for viewers to derive utility from the following behavior. Usually, emotionally abundant voices have prosody variations that are expressive in nature, and the viewers need fewer cognitive resources to process

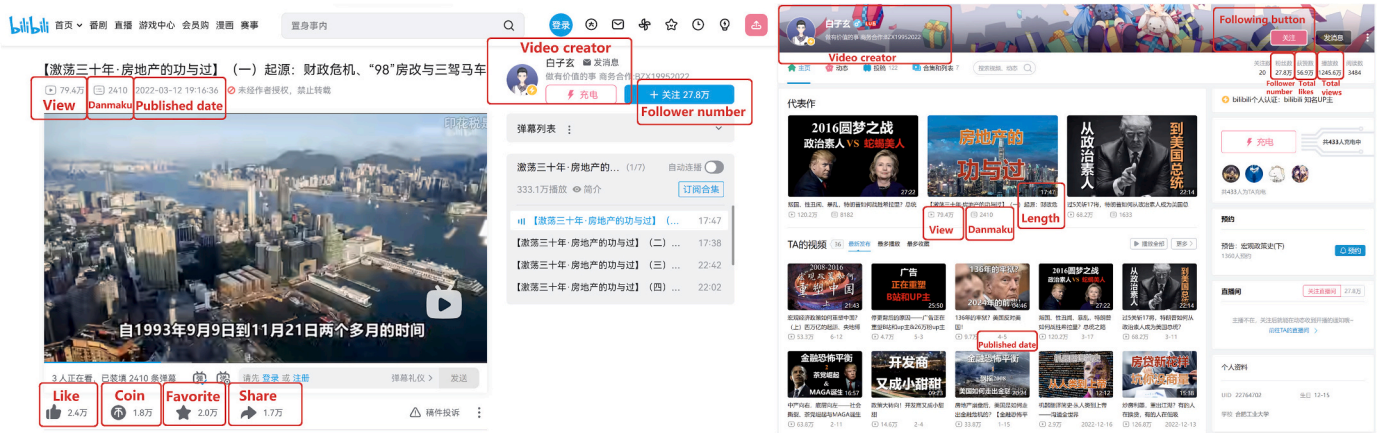


Fig. 1. Screenshots of a user-generated video and the homepage of the video creator on Bilibili.

acoustic information with higher loudness variability [72]. In other words, the higher loudness variability of a video creator, the fewer mental efforts viewers need to gain benefits from the video content [73]. Thus, viewers could consistently accrue knowledge and insights from the video creator in a long run. Accordingly, we propose the following hypothesis:

H3. Loudness variability of a video creator is positively related with the change in the number of followers he/she receives.

Although high vocal pitch is tempting and might temporarily attract viewers, the constant strident voice may dampen viewers' cognition of the focal content [74]. High-pitch videos are just like a sweet-and-tasty meal that offers short-term gratification but eventually becomes bored with. Constant strident voice might be a trigger of annoyance [75]. Viewers might unfollow the creators if they perceive annoyance from the continuous interactions [15]. Hence, it is readily understandable why the video creator would elicit a negative utility when presenting with a high pitch, particularly considering viewers' anticipation for superior video content accompanied by soothing tones. Despite media content with bright lights or harsh sounds are easier to arouse individuals' immediate pleasure, quality content with soothing sound ensures continuous recognition in the long run [12]. Accordingly, we propose the following hypothesis:

H4. Vocal pitch of a video creator is negatively related with the change in the number of followers he/she receives.

3.2. The effect of quantity signals

During the decision-making process, acoustic signals and quantity signals may complement each other. For example, it has been found that interviewees' LinkedIn information (textual cues) supplements interviewers' first-stage audio evaluation (verbal cues) of job candidates through field experiments [76]. The interactive effect between the content and content creator has been validated to influence the virality of a post in a traditional text-based social media context [14].

Studies have indicated that variations in loudness can impact viewers' perceptions and emotional responses [65]. Dynamic changes in audio levels can evoke heightened attention, arousal, and emotional impact, making the viewing experience more engaging and memorable. Videos with higher length could corroborate the positive effect of loudness variability on viewer engagement by providing abundant content to satisfy viewers' content needs [56]. When video creators produce longer videos, there is more room to reflect dynamic changes in

the audio levels and sound effects. These variations can help create a more immersive and engaging experience for viewers, capturing their attention and enhancing their overall viewing experience [56,77]. As a result, when viewers encounter these fluctuations in loudness while watching a longer video, they are more likely to have a pleasant experience and like it. Accordingly, we propose the following hypothesis:

H5a. The average length of videos strengthens the positive effect of loudness variability on the number of likes a video creator receives.

It should be noted that viewers may also experience attention depletion [78], especially when videos are lengthy and with harsh high-pitched voices [28]. When a video is protracted, viewers are overloaded with tedious information. In this case, viewers would shift their focus to the efficiency in delivering a specific message and prioritize brevity and conciseness of a video with limited attention paid to the vocal performance [79]. Consequently, the number of likes given by viewers for a longer video may decrease correspondingly, and the positive influence of vocal pitch variations on viewer engagement may be overshadowed by other factors such as content relevance, visual appeal, or overall production quality. Accordingly, we propose the following hypothesis:

H5b. The average length of videos weakens the positive effect of vocal pitch on the number of likes a video creator receives.

Video creators with more published videos are expected to be more productive and active [80]. A high number of published videos at least signals that the creator is studious in continuously outputting valuable content. These video quantity signals, such as the number of weekly published videos, have been identified as influential in terms of the number of followers a video creator can attract in the online context [13,15]. When the number of weekly published videos of a video creator is high, viewers will have an anticipation of quality content production from the focal creator in a long run [81], and regard the quantity signals as a proxy of ongoing benefits. When the loudness variability of a video creator motivates viewers' following behaviors, the number of weekly published videos further strengthens this positive effect by serving as a quantity signal and attracting more followers. Accordingly, we propose the following hypothesis:

H6a. The number of weekly published videos strengthens the positive effect of loudness variability on the change in the number of followers a video creator receives.

Previous research has revealed a potential negative relationship between the vocal pitch of a video creator and viewers' evaluation of the

The Procedure of Data Collection and Processing

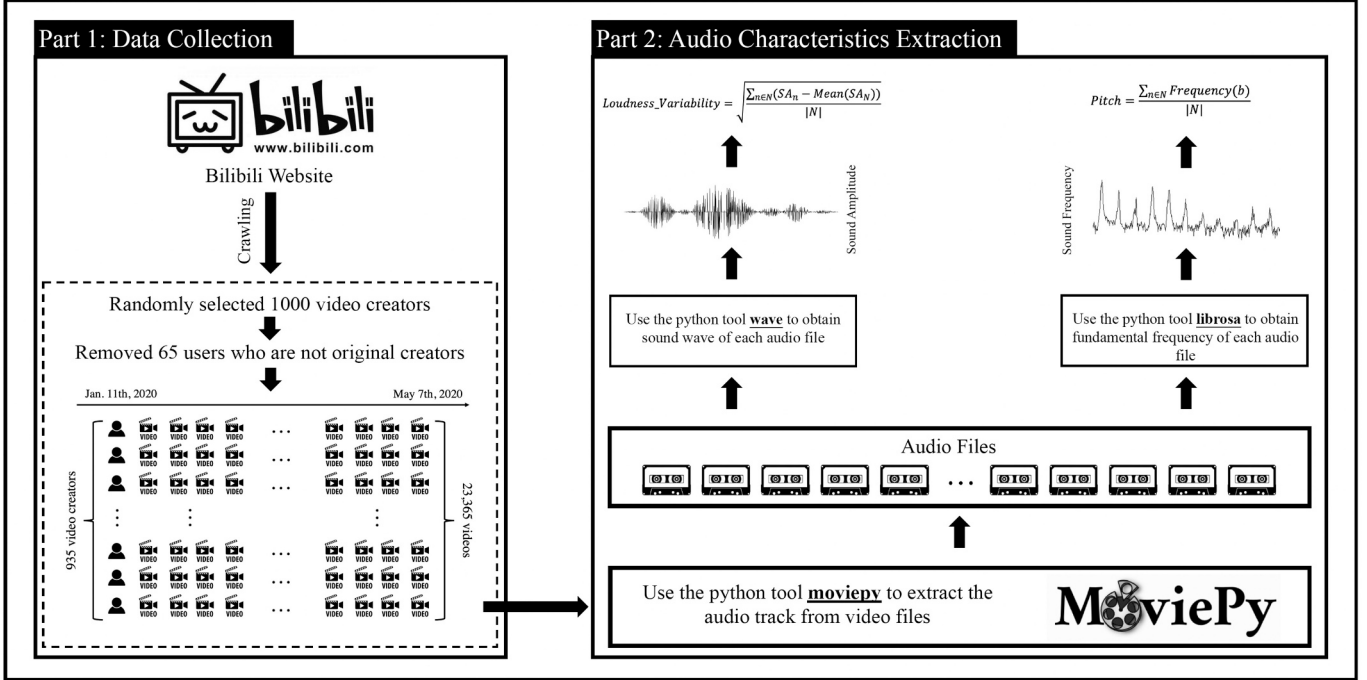


Fig. 2. The procedure of data collection and processing

videos [19]. However, a higher number of published videos may lead to an expanded array of new content for viewers. This consistent provision of new content allows creators to maintain sustained audience attention [82], thereby diminishing the negative impact of vocal pitch on the change in the number of followers. Besides, the compounding effect of diverse videos from the focal creator enables viewers to divert their attention on other things, e.g., information quality, information relevance, to their specific interests, thus mitigating the negative impact brought by the variations in vocal pitch. Accordingly, we propose the following hypothesis:

H6b. The number of weekly published videos weakens the negative effect of vocal pitch on the change in the number of followers a video creator receives.

4. Research methodology

4.1. Data acquisition and preprocessing

The data was collected from Bilibili, one of the largest online user-generated video platforms in China. On Bilibili, video creators can publish their self-created videos to share thoughts, feelings, and expertise. Viewers could follow a video creator per their interests. The videos from the video creator can also be liked, shared, coined and favorited by other viewers. Other information, such as the length of each video and the total number of videos from a creator, are displayed on the homepage. Normally, viewers will follow a creator after browsing the creator's homepage. The screenshots of a typical user-generated video on Bilibili and the homepage of the video creator are shown in Fig. 1.

Fig. 2 below shows the data collection and processing procedure of this paper. All the videos uploaded from January 11, 2021, to May 7,

Table 1
Variables and measures.

Variable	Measure
$Like_{i,t}$	The incremental number of likes received by all the videos published by creator i in week t .
$FollowerChangeR_{i,t}$	The incremental ratio of followers for creator i between week t and week $t-1$ comparing with the number of followers in week t .
$FollowerChangeD_{i,t}$	If the incremental number of followers for creator i between week t and week $t-1$ is larger than 0, the value equals to 1; otherwise, the value is 0.
$LoudnessVariability_{i,t}$	The average standard deviation of sound amplitude of creator i among all the videos published in week t .
$VocalPitch_{i,t}$	The average number of fundamental voice frequency of creator i among all the videos published in week t .
$Length_{i,t}$	The average length of videos published by creator i in week t .
$VideoCount_{i,t}$	The average number of videos published by creator i in week t .
$ContentQuality_{i,t}$	The mean difference between the number of positive and negative words contained in the comments received by all the videos published by creator i in week t .
$View_{i,t}$	The incremental number of views received by all the videos published by creator i in week t .
$LikeRatio_{i,t}$	The number of likes ($Like_{i,t}$) divided by the number of views ($View_{i,t}$) received by all the videos published in week t .
$Speed_{i,t}$	The average number of words contained per minute in the videos published by creator i in week t .
$Zcr_{i,t}$	The average number of times the voice crosses the zero axes among all the videos published by creator i in week t .
<i>Category dummies</i>	11 dummy variables explaining the specific category of videos published by creator i in week t . For each categorical variable, 1 denotes a video pertains to the focal category; otherwise 0.
$Tenure_{i,t}$	The number of days since creator i published the first video in week t .
$StartFollower_{i,t}$	The total number of followers creator i has before week t .
$TotalVideo_{i,t}$	The total number of videos creator i has published before week t .

2021 and related information were collected using API and Python packages. Among them, observations from 935 video creators across 25 weeks were identified. But it should be noted that not all the video creators release videos every week. If there was not any video uploaded in a week, the observation value for that week would be 0. After teasing out all the 0 values, we finally had 18,307 observations at the creator-week level covering five video areas: gourmet, game, daily life, knowledge, and fashion. These five areas fall into the scope of our investigation because the associated videos are generally user-generated videos.

4.2. Variables

All the variables are aggregated at the creator-weekly level. i denotes each video creator in the video sharing platform, and t denotes each week. There are two dependent variables in this paper. The first dependent variable (to test H1, H2, and H5), $FollowerChangeR_{i,t}$ captures the incremental ratio of followers for video creator i between week t and week $t-1$ comparing with week $t-1$. The second dependent variable (to test H3, H4, and H6), $Like_{i,t}$, is the number of likes received by the videos published by creator i in week t . Table 1 exhibits the measurements of variables.

The two independent variables are loudness variability ($LoudnessVariability_{i,t}$) and vocal pitch ($VocalPitch_{i,t}$) of a video creator. Loudness variability is measured by the average standard deviation of sound amplitude of the released videos, while vocal pitch gives the fundamental frequency of a speech, indicating the “harshness” of a given sound. We also calculated the speed rate of the video creator as a sign of voice tempo and the zero-crossing rate of the published videos in each week to control the influences of other sound-related characteristics.

The measurement of acoustic characteristics is calculated through speech recognition analytics. In detail, we first extracted the audio data from video files using the moviepy, a python tool. Then we used the spleeter package to distinguish vocals and accompaniment separation, eliminating the possible influences of background sounds. After that, *wave* and *librosa*, were applied to obtain the sound amplitude and sound

frequency information, respectively. Finally, we calculated loudness variability and pitch as follows:

$$Loudness_Variability = \sqrt{\frac{\sum_{n \in N} (SA_n - Mean(SA_N))^2}{|N|}}$$

where n is a particular audio frame; N is the set of audio frames; $|N|$ is the number of audio frames; SA denotes the sound amplitude of an audio frame.

$$Vocal_Pitch = \frac{\sum_{n \in N} Frequency(b)}{|N|}$$

where b is a certain bin; $Frequency(b)$ represents the fundamental frequency (F0) of b , $X_n(b)$ represents the weighted value of b for the audio frame n .

Besides, we included two quantity signals to test their moderation effects. The number of weekly published videos, is operationalized as the number of videos published by creator i in week t (denoted as $VideoCount_{i,t}$). Average length of videos captures the average length of videos published by creator i in week t (denoted as $Length_{i,t}$).

4.3. Alternatives

Alternative explanations on the dependent variables were controlled as well. First, considering the presence of heteroscedasticity in the White test, we employed clustering for robust standard errors. Second, other acoustic features in the video were controlled, including speed (denoted as $Speed_{i,t}$) and zero-crossing rate (denoted as $Zcr_{i,t}$). Third, video creators' past experience was controlled by including $Tenure_{i,t}$, the number of days since creator i published the first video in week t . The baseline number of followers before week t (denoted as $StartFollower_{i,t}$) and the total number of published videos before week t (denoted as $TotalVideo_{i,t}$) of the focal creator were also controlled. Forth, the quality of video content, denoted as $ContentQuality_{i,t}$, was controlled by calculating the number of positive words minus the number of negative words in the

Table 2
Effects of acoustic characteristics.

	Column 1	Column 2	Column 3	Column 4
	DV = Ln ($Like_{i,t} + 1$)		DV = Ln ($FollowerChangeR_{i,t} + 1$)	
Ln($LV_{i,t} + 1$)	0.334*** (6.05)	0.314*** (6.75)	0.0001** (1.84)	0.0001** (1.69)
Ln($PI_{i,t} + 1$)	0.307*** (4.49)	0.291*** (3.98)	-0.0002** (-1.92)	-0.0001** (-1.78)
Ln($Length_{i,t} + 1$)	0.081** (2.12)	0.098** (2.48)	0.0001 (1.34)	0.0001 (1.28)
Ln($VideoCount_{i,t} + 1$)	-0.019 (-0.69)	-0.021 (-0.63)	0.002** (2.11)	0.002** (2.03)
Ln($Length_{i,t} + 1$) * Ln($LV_{i,t} + 1$)		0.034*** (2.92)		
Ln($Length_{i,t} + 1$) * Ln($PI_{i,t} + 1$)		-0.044** (-2.51)		
Ln($VideoCount_{i,t} + 1$) * Ln($LV_{i,t} + 1$)				0.0002* (1.91)
Ln($VideoCount_{i,t} + 1$) * Ln($PI_{i,t} + 1$)				0.0003* (1.75)
Control variables	Included	Included	Included	Included
Constant	3.143** (2.22)	3.158** (2.28)	0.003*** (4.92)	0.002*** (4.51)
Adjusted R-squared	0.576	0.584	0.162	0.163
Observation	18,307	18,307	18,307	18,307
Week-fixed effect	YES	YES	YES	YES
Individual-fixed effect	YES	YES	YES	YES
Category dummies	YES	YES	YES	YES

Note: Robust T-statistics in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3
Results with alternative measures of dependent variables.

	DV = <i>LikeRatio</i> _{i,t}			DV = <i>FollowerChange</i> _{i,t}	
	Column 1	Column 2		Column 3	Column 4
$\ln(LV_{i,t} + 1)$	0.002** (2.06)	0.002** (2.22)	$\ln(LV_{i,t} + 1)$	0.092** (2.06)	0.083** (2.04)
$\ln(PI_{i,t} + 1)$	0.002** (2.13)	0.002** (1.98)	$\ln(PI_{i,t} + 1)$	-0.087* (-1.91)	-0.085* (-1.72)
$\ln(Length_{i,t} + 1)$	0.002*** (3.21)	0.002*** (2.87)	$\ln(VideoCount_{i,t} + 1)$	0.461** (2.51)	0.419** (2.33)
$\ln(Length_{i,t} + 1) * \ln(LV_{i,t} + 1)$		0.002** (2.12)	$\ln(VideoCount_{i,t} + 1) * \ln(LV_{i,t} + 1)$		0.055*** (2.41)
$\ln(Length_{i,t} + 1) * \ln(PI_{i,t} + 1)$		-0.003*** (-2.66)	$\ln(VideoCount_{i,t} + 1) * \ln(PI_{i,t} + 1)$		0.015* (1.67)
Control variables	Included	Included	Control variables	Included	Included
Constant	-13.013*** (-5.48)	-12.001*** (-5.09)	Pseudo R ²	0.073	0.073
Adjusted R-squared	0.058	0.059	Log-likelihood	-4640.3435	-4603.7341
Observation	18,307				
Week-fixed effect	YES				
Individual-fixed effect	YES				
Category dummies	YES				

Note: Robust T-statistics in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

comments through linguistic inquiry and word count (LIWC) techniques. Last, to ensure the consistency of video content and corresponding audio, the audio of each video was converted into text in the first place, then “jieba” word segmentation and “HanLP” speech tagging were applied and LDA topic modelling was conducted based on the title, description, and text of audio content [83]. Following Blei et al.'s work [83], the highest coherence score was set as the number of topics (K), where $K = 11$. Thus, the 11 topics-related categorial variables generated through LDA, including “Competition”, “Video Game”, “Beauty”, “International business”, “Drama”, “Interview”, “Family”, “Daily life”, “Gourmet”, “Story”, and “Table Game”, were incorporated into the regression model as dummy variables (see Appendix A1 for detailed information). For each variable, 1 denotes a video creator belongs to the focal category, and 0 otherwise.

In Appendix A2, Table A2-1 presents the descriptive statistics of key variables. Besides, multi-collinearity was tested in the data. The VIF values are ranged from 1.07 to 4.73, suggesting that multi-collinearity should not be an issue in this paper. Table A2-2 presents the correlations between variables.

5. Data analysis and results

5.1. Empirical specification

To examine the effect of acoustic characteristics on the number of likes, the following model is adopted:

$$\begin{aligned} \ln(Like_{i,t} + 1) = & \ln(LV_{i,t} + 1) + \ln(PI_{i,t} + 1) + VoiceControls_{i,t} \\ & + VideoControls_{i,t} + \sum (Category\ Dummies_i) + \alpha_i + \gamma_t \\ & + \epsilon_{i,t} \end{aligned}$$

Logarithmic (log) values are used following the general practice since the variables are skewed [51,84]. $\ln(Like_{i,t})$ is the log-transformed number of likes received by creator i in week t . Individual fixed effect, α_i , is included to capture the unobserved time-invariant differences across video creators. Week fixed effect, γ_t , is also included. Category dummy variables were included in the model to account for the effects of video contents. In addition, other video attributes (i.e., $\ln(Zcr_{i,t} + 1)$, $\ln(Speed_{i,t} + 1)$, $\ln(StartFollower_{i,t} + 1)$, $\ln(TotalVideo_{i,t} + 1)$, $\ln(View_{i,t} + 1)$, $\ln(Tenure_{i,t} + 1)$, and $\ln(ContentQuality_{i,t})$) are controlled. Based on Hausman test, a fixed-effect model is more appropriate than a random-

effect model ($\chi^2 = 83.75$, $p < 0.01$) for estimation.

Then, to examine the effect of acoustic voice characteristics on the change in the number of followers, the following model is adopted:

$$\begin{aligned} \ln(FollowerChangeR_{i,t} + 1) = & \ln(LV_{i,t} + 1) + \ln(PI_{i,t} + 1) \\ & + VoiceControls_{i,t} + VideoControls_{i,t} \\ & + \sum (Category\ Dummies_i) + \alpha_i + \gamma_t + \epsilon_{i,t} \end{aligned}$$

$\ln(FollowerChangeR_{i,t} + 1)$ is the log-transformed incremental ratio of followers for creator i between week t and week $t-1$ comparing with week $t-1$. Individual fixed effect, α_i , is included to capture the unobserved time-invariant differences across video creators. Week fixed effect, γ_t , is also included. Category dummy variables were included in the model to account for the effects of video contents. In addition, other video attributes (i.e., $\ln(Zcr_{i,t} + 1)$, $\ln(Speed_{i,t} + 1)$, $\ln(TotalVideo_{i,t} + 1)$, $\ln(Tenure_{i,t} + 1)$, $\ln(View_{i,t} + 1)$, $\ln(ContentQuality_{i,t})$) are controlled. Based on Hausman test, a fixed-effect model is more appropriate than a random-effect model ($\chi^2 = 98.34$, $p < 0.01$) for estimation.

5.2. The effect of acoustic and quantity cues

Table 2 presents the results of fixed-effect regression using panel data with both individual effects fixed and time effects fixed. Column 1 shows the direct impact of loudness variability and vocal pitch on the number of likes. Supporting H1 and H2, both loudness variability ($\beta = 0.334$, $p < 0.01$) and vocal pitch ($\beta = 0.307$, $p < 0.01$) have a significantly positive impact on the number of likes. Column 2 shows the average length of videos positively moderates the relationship between loudness variability and the number of likes ($\beta = 0.034$, $p < 0.01$), thus H5a is supported. Also, the average length of videos positively moderates the relationship between vocal pitch and the number of likes ($\beta = -0.044$, $p < 0.05$), supporting H5b. It means that a video creator with a higher pitch voice will receive more likes when the video is longer but receive fewer likes when the video is shorter.

Column 3 exhibits the benchmark model on the change in the number of followers with both individual- and time-fixed effects. The results indicate that loudness variability is positively related with the increase in the number of followers ($\beta = 0.0001$, $p < 0.05$). Thus, H3 is supported. But vocal pitch negatively affects the changes in the number of followers ($\beta = -0.0002$, $p < 0.05$), supporting H4. Column 4 confirms that the number of weekly published videos positively moderates the

relationship between loudness variability and the change in the number of followers ($\beta = 0.0002, p < 0.1$), thus H6a is supported. It tells that the positive impact of voice variability on the increase of followers is stronger for video creator with a higher number of weekly published videos. Column 4 further confirms the positive effect of interaction term, and reveals that the number of weekly published videos negatively moderates the negative relationship between vocal pitch and the change in the number of followers ($\beta = 0.0003, p < 0.1$), thus H6b is supported.

5.3. Robustness check

To check the robustness of the main results, alternative measures of dependent variables were adopted. The results are summarized in Table 3. Considering the difference in the exposure size of each video, we construct a proportional measure of likes, $LikeRatio_{i,t}$, which is the number of likes ($Like_{i,t}$) divided by the numbers of views ($View_{i,t}$) for creator i in week t . Fixed-effect models were regressed on the logarithmic values of $LikeRatio_{i,t}$. Moreover, a discrete variable, denoted by $FollowerChangeD_{i,t}$ (where 1 indicates that the incremental number of followers for creator i in week t is larger than 0; otherwise 0), was used to measure the changes in the number of followers, as shown in column 3 and 4. The relationships between the acoustic characteristics of video creators and the dependent variables are aligned with the main findings.

Several other robustness checks were conducted to further validate the findings, including sub-sample analysis, video-level analysis and alternative measurement intervals of dependent variables. The details are shown in the Appendix A3, A4 and A5 respectively. It can be told that our results are quite robust.

6. Discussion

6.1. Summary of results

To improve the performance of user-generated videos, this paper investigates the impact of acoustics characteristics on the two key performance indicators of a focal creator based on data from Bilibili. The results reveal that loudness variability of the video creator significantly increases the number of likes a video creator receives, and exerts a positive effect on the change in the number of the creator's followers. In contrast, vocal pitch has a significant positive impact on the number of likes but a significant negative impact on the change in the number of followers.

Moreover, the moderation effect of video quantity signals on the relationship between acoustic characteristics and video performance is confirmed. Specifically, the average length of videos positively moderates the relationships between loudness variability and the number of likes but negatively moderates the impact of vocal pitch on the number of likes. In contrast, the number of weekly published videos positively moderates the relationship between loudness variability and the change in the number of followers, and positively moderates the negative impact of vocal pitch on the change in the number of followers.

6.2. Theoretical contributions

Leveraging signaling theory of portfolios, this work contributes to the literature in several important aspects. First, existing research on user-generated videos have examined the attributes of video creators' performance from the perspective of creators' characteristics, quality of video content, and interactive relationships [15,41]. Compared with other forms of user-generated content, videos are more complex and full of audio data, which are important in influencing viewers' decision-making [16,85]. Our paper adds to the literature by using an acoustic lens [26] to conceptualize and extend the influencing factors of video creators' performance.

Second, current literature on acoustic decision-making focused on the cognitive or emotional fit between audio contents and brand images

[21,86], specific acoustic characteristics were scarcely explored. Although there are some explorations on acoustic characteristics, they mostly used small-scale experiments for data acquisition which are not easily to be generalized to a broad context [19,64,87]. This paper abstracts vocal features of videos using automated speech recognition analytics from Bilibili. The uniqueness of our dataset enables large-scale longitudinal investigation of the acoustic characteristics of video creators.

Third, our results provide empirical evidence on the heterogeneous impacts of acoustic characteristics of video creators on the number of likes and the change in the number of followers. As documented in the previous literature, viewers' liking behavior is a manifestation of their short-term engagement, while the following behavior reflects a long-term engagement [10,39]. On this basis, this paper contributes to the literature by examining the plausible antecedents of both short-term and long-term engagement on user-generated video platforms [15,39,38].

Moreover, leveraging the theory of signal portfolios [29,30], this paper reveals the underpinning mechanisms between acoustics cues, quantity cues, and video creators' online performance. The main results corroborate that both acoustic and quantity signals influence viewers' perceptions and behaviors [88]. But it should also be noted that, based on the sub-sample analysis, the portfolios of acoustic signals and quantity signals lose effect for female creators regarding to viewers' liking decisions, and it remains the same for male creators regarding to viewers' following decisions. These findings advance the understanding of the heterogeneous impacts of genders on video creators' performance.

Last, it is found that when the average length of videos is low, creators' high-pitched voices could have more likes from viewers, and the number of weekly published videos could weaken the negative effect of vocal pitch on the change in the number of followers. The reason is because, as the videos get longer or more frequent exposure, viewers tend to lose interest in those high-pitched videos. These findings contribute to existing literature by empirically validating that long videos do not always lead to better performance [77].

6.3. Practical implications

This paper also has several decision-support implications to practitioners in the video-sharing industry. To begin with, the results indicate that loudness variability is beneficial for gaining likes and followers. Consequently, video creators can take advantage of their loudness variability to improve their performance. However, video creators should be cautious in modulating vocal pitch as a higher pitch in the video is negatively related with the increase in the number of followers. Video creators with a higher number of likes but with a lower number of followers can adjust their voicing strategies by lowering their voices. Since vocal pitch is negatively related with the number of likes when the video is long, video creators should only adopt high-pitch voices for short-framed videos to attract more likes.

Besides, inspired by the impact of quantity signals, video sharing platform operators should thus be concerned not only about video creators' acoustic characteristics but also adopt a more comprehensive strategy. For instance, platform operators could offer incentives to encourage video creators to release more videos apart from modulating acoustic characteristics. Because the number of weekly published videos could strengthen the positive impact of loudness variability and weaken the negative impact of vocal pitch on the change in the number of followers.

Moreover, our results could inspire better design of platforms to encourage sharing of user-generated videos. For example, considering the average length of videos would strengthen the positive impact of loudness variability and weaken the positive impact of vocal pitch on the change in the number of followers, platform designers could encourage video creators to produce relatively long videos when their loudness variability is consistently high. Specifically, platform operators should be cautious in designing respective tactics for male and female video

creators based on the sub-sample analysis results.

6.4. Limitations and future research

Admittedly, our work still has some limitations. First, this paper only involves a single user-generated video platform (i.e., Bilibili) in a fixed timeframe, so caution must be exercised in generalizing the findings to other video sharing platforms. Further research should be undertaken to investigate a wider range of user-generated video platforms covering more samples. Causal analysis using other research methods, such as laboratory experiments, is also worth further exploration.

Second, we acknowledge that video creators' external status capital (e.g., a celebrity from other platforms) and viewers' characteristics (e.g., individuals' tendency to follow an account) may also influence the change in the number of followers on the focal platform. In addition, other acoustic characteristics, such as speaking styles of video creators and the consistency between acoustic characteristics and video contents, may also be important for viewers' overall perceptions of a video. Thus, further explorations into these aspects could offer better understanding of the relationships between acoustic characteristics and video creators' performance.

Last, the values of loudness variability and vocal pitch of each video creator were aggregated at a weekly level in the main results. However, the choice of week as an analysis unit may not fully capture the dynamic variations in the voice characteristics of a video creator. Also, since viewers' demographic information is not included in the current data, the unobserved individual characteristics may bias the results. Future research could make these up using fine-grained data and advanced analytics.

7. Conclusion

Based the signaling theory of portfolios, this paper advances the understanding of video creators' performance by theorizing a research model depicting the relationships between acoustic characteristics, signals of video quantity, and performance metrics manifested by the number of likes and the change in the number of followers a video creator receives. This work empirically investigates how the voice of video creators influences their performance on video sharing platforms as well as the boundary conditions (i.e., the average length of videos and

the number of weekly published videos). According to the results, video creators will receive a higher number of likes (short-term engagement) when loudness variability and vocal pitch are both high. In contrast, the increase in the number of their followers (long-term engagement) is related with higher loudness variability and lower vocal pitch.

CRedit authorship contribution statement

Shixuan Fu: Writing – review & editing, Writing – original draft, Conceptualization. **Yan Wu:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Qianzhou Du:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Chenwei Li:** Writing – review & editing, Conceptualization. **Weiguo Fan:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Qianzhou Du reports financial support was provided by National Natural Science Foundation of China. Qianzhou Du reports financial support was provided by National Social Science Foundation of China. Shixuan Fu reports financial support was provided by National Natural Science Foundation of China. Chenwei Li reports financial support was provided by Xi'an Jiaotong-Liverpool University.

Data availability

Data will be made available on request.

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Appendix A1: Video content calculation

To avoid the confounding effects of video content, topic modelling analysis was conducted using Latent Dirichlet Allocation (LDA) model on the content of videos (e.g., title, video description, and textual information of audios). Specifically, Baidu API was applied to convert the audios into textual content. Second, we used the “HanLP” package [89] in Python for Chinese text tokenization. We also removed the meaningless words (so-called “stop words”) in the natural language processing. Then, Part-of-Speech (POS) technique was applied to pre-process the textual content and to train the LDA model based on the nouns in the textual content. To determine the topic number k , we followed Griffiths and Steyvers [90] and Blei et al. [83] by calculating the coherence scores of LDA models with different topic numbers $k \in K = \{1, 2, 3, 4, \dots, 15\}$ as shown in Fig. A1. We set 11 as the best topic number because the LDA model has the highest coherence score when $K = 11$, ensuring the best consistency of content. Table A1 shows the detailed keywords of the 11 topics, which are “Competition”, “Video Game”, “Beauty”, “International business”, “Drama”, “Interview”, “Family”, “Daily life”, “Gourmet”, “Story”, and “Table Game”.

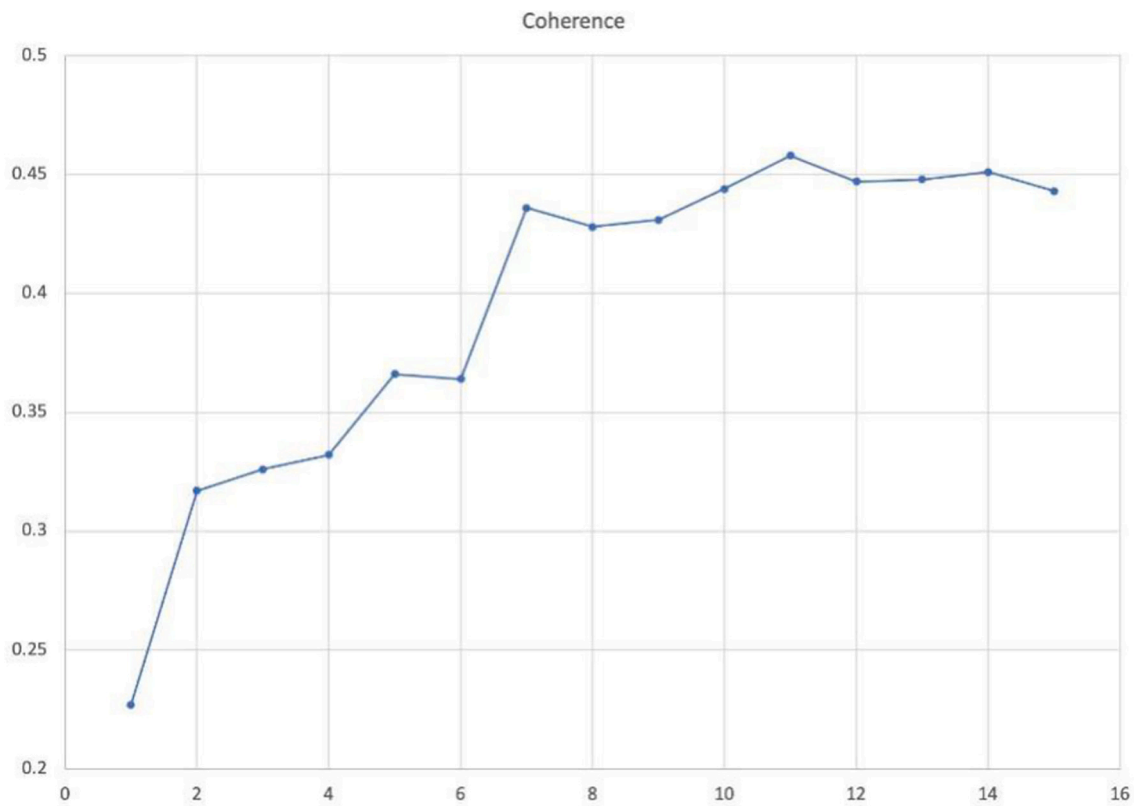


Fig. A1. Coherence scores for different number of topics.

Table A1
Representative words of the 11 topics.

Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9	Topic 10	Topic 11
Competition	Video game	Beauty	International business	Drama	Interview	Family	Daily life	Gourmet	Story	Table game
Competition	Game	Color	China	Kid	Question	Mother	Now	Smell	Now	Player
Skill	Player	Appearance	American	Story	Profession	Father	Life	Taste	Childhood	Police
Hero	Plot	Eye	Firm	Human	Project	Sister	Work	Delicious	Today	Good characters
Position	Task	Skin	Market	Movie	Method	Brother	Friend	Beef	Man	Speech
Opponent	Role	Clothes	Country	Actress	Work	Son	Classmate	Flavor	Place	Identity
Contestant	Weapon	Hair	Economy	Mother	Design	Kid	Sport	Egg	Night	Logic
Team	Map	Body	Industry	History	Study	Wife	Today	Pepper	House	Number
Advantage	Mode	Muscle	Currency	Life	Company	Grandpa	Parents	Chili	Boss	Clue
Coach	Enemy	Face	Product	Actor	Techniques	Grandma	Exam	Scent	Tomorrow	Murder
Rival	Fight	Sculpt	Transaction	Event	Interview	Baby	Kid	Seafood	Yesterday	Probability

Appendix A2: Descriptive statistics and correlations

Table A2-1
Descriptive statistics of variables.

Variable	Observation	Mean	Standard deviation	Min	Max
Like	18,307	5924.13	2850.42	0	615,746
FollowerChangeR	18,307	0.005	0.026	−0.924	1.924
FollowerChangeD	18,307	0.831	0.375	0	1
LoudnessVariability	18,307	4104.7	2340.329	27.929	17,891.5
VocalPitch	18,307	279.951	162.023	76.152	2205

(continued on next page)

Table A2-1 (continued)

Variable	Observation	Mean	Standard deviation	Min	Max
Length	18,307	1217.958	2508.341	3.12	6538.768
VideoCount	18,307	2.399	2.426	1	33
ContentQuality	18,307	3.455	4.298	−35	100
View	18,307	115,477.3	305,169.3	3	6,589,119
LikeRatio	18,307	0.052	0.022	0	1
Speed	18,307	215.919	173.129	0.01	1637.96
Zcr	18,307	2222.401	4079.779	6.530	332,154.4
Tenure	18,307	1056.842	642.413	13	4379
StartFollower	18,307	445,945.7	756,177	425	7,252,954
TotalVideo	18,307	321.452	435.183	1	6013

Table A2-2

Correlations between key variables.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.FollowerChangeR	1.0000													
2.FollowerChangeD	0.6046	1.0000												
3.Like	0.5358	0.1575	1.0000											
4.Speed	0.1699	0.2619	0.1777	1.0000										
5.LoudnessVariability	0.1356	0.2755	0.1414	0.5297	1.0000									
6.VocalPitch	−0.1417	−0.3040	0.1687	0.4619	0.5328	1.0000								
7.Tenure	−0.0621	−0.1553	0.0240	−0.0626	−0.0738	−0.1073	1.0000							
8.VideoCount	0.0935	0.2175	0.1570	0.3727	0.4462	0.4346	−0.1018	1.0000						
9.TotalVideo	0.0147	0.0279	0.1006	0.1573	0.1658	0.1586	0.2337	0.4004	1.0000					
10.StartFollower	0.4131	0.1269	0.5333	0.1863	0.1325	0.1440	0.1352	0.0830	0.1441	1.0000				
11.Zcr	0.0512	0.1177	0.0598	0.2329	0.2799	0.2754	−0.0529	0.2161	0.0774	0.0563	1.0000			
12.ContentQuality	0.0434	0.1689	0.0294	0.3127	0.3719	0.3830	−0.0761	0.2737	0.0599	0.0304	0.1799	1.0000		
13.Length	0.0763	0.1275	0.0879	0.2253	0.1948	0.2632	0.0009	0.5580	0.2400	0.0766	0.1215	0.1764	1.0000	
14. View	0.1869	0.1195	0.6457	0.1113	0.0317	−0.0121	0.0791	0.1339	0.0221	0.6261	0.0082	0.0902	0.1407	1.0000

Notes: All the coefficients are significant at the 0.01 level (2-tailed).

To illustrate the reasons why other performance indicators, such as the number of shares, the number of favorites, and the number of coins, are not included in the model, [Table A2-3](#) reports the high correlations between the number of likes and these variables.

Table A2-3

Correlations between potential dependent variables.

	Mean	Standard deviation	FollowerChangeR	Share	Favorite	Coin	Like
FollowerChangeR	0.005237	0.025777	1.0000				
Share	313.7658	2112.181	0.1082	1.0000			
Favorite	1119.751	4937.027	0.1559	0.6108	1.0000		
Coin	2012.66	11,415.52	0.1016	0.5008	0.7797	1.0000	
Like	4235.606	19,531.72	0.1182	0.6556	0.7448	0.8126	1.0000

Notes: All the coefficients are significant at the 0.01 level (2-tailed).

Appendix A3: Sub-sample analysis

Sub-sample analysis was performed for male and female video creators as shown in [Table A3](#). There is no much difference with the main results in terms of gender effect on the relationships between loudness variability, vocal pitch and the dependent variables. However, it is interesting to note that the moderation effect of the average length of weekly published videos only holds for male video creators, while the moderation effect of the number of weekly published videos only holds for female video creators. Therefore, male creators should be more cautious about the use of video quantity signals if they want to earn more likes. Female creators should be cautious as well if they want to have more followers.

Table A3

Sub-sample analysis.

	DV = $\ln(\text{Like}_{i,t} + 1)$				DV = $\ln(\text{FollowerChangeR}_{i,t} + 1)$			
	Male		Female		Male		Female	
$\ln(LV_{i,t} + 1)$	0.457*** (3.52)	0.419*** (2.95)	0.185* (1.92)	0.166* (1.74)	0.0001 (0.31)	0.0001 (0.25)	0.0001*** (2.38)	0.0001* (1.86)
$\ln(PI_{i,t} + 1)$	0.530*** (2.93)	0.521*** (2.90)	0.053** (2.17)	0.045** (1.99)	−0.0002* (−1.67)	−0.0002* (−1.67)	−0.001** (−2.01)	−0.001** (−2.12)
$\ln(\text{Length}_{i,t-1})$	0.174** (2.11)	0.157* (1.92)	0.066** (2.01)	0.124* (1.93)	0.0001 (0.33)	0.0001 (0.26)	0.0001 (0.24)	0.0001 (0.21)
$\ln(\text{VideoCount}_{i,t-1})$	−0.353*** (−2.84)	−0.366*** (−2.92)	0.093 (0.51)	0.088 (0.44)	0.0001* (1.84)	0.0001* (1.82)	0.0001*** (2.58)	0.0001** (2.54)
$\ln(\text{Length}_{i,t} + 1) * \ln(LV_{i,t} + 1)$		0.071* (1.84)		−0.037 (−0.68)				

(continued on next page)

Table A3 (continued)

	DV = Ln (Like _{i,t} + 1)				DV = Ln(FollowerChangeR _{i,t} + 1)			
	Male		Female		Male		Female	
Ln(Length _{i,t} + 1)* Ln(PI _{i,t} + 1)		−0.062*		−0.357				
		(−1.73)		(−0.31)				
Ln(VideoCount _{i,t} + 1)* Ln(LV _{i,t} + 1)						0.0001		0.0001*
						(0.58)		(1.77)
Ln(VideoCount _{i,t} + 1)* Ln(PI _{i,t} + 1)						0.0001		0.0001*
						(0.88)		(1.68)
Control variables	Included	Included	Included	Included	Included	Included	Included	Included
Constant	−15.883***	−15.787***	−5.832**	−5.282**	0.002***	0.002***	0.033**	0.033**
	(−8.57)	(−8.54)	(−1.98)	(−2.01)	(2.81)	(2.77)	(1.85)	(1.88)
Adjusted R-squared	0.552	0.598	0.554	0.589	0.088	0.088	0.112	0.114
Category dummies	YES	YES	YES	YES	YES	YES	YES	YES
Observation	18,307	18,307	18,307	18,307	18,307	18,307	18,307	18,307
Week-fixed effect	YES	YES	YES	YES	YES	YES	YES	YES

Note: Robust T-statistics in parentheses, *** p < 0.01, ** p < 0.05, * p < 0.1.

Appendix A4: Video-level analysis

We further conducted video level analysis with creator and category dummies. Other possible influencing factors were controlled, including other acoustic features in the video, i.e., speed and zero-crossing rate. Content quality and the number of views were controlled as well. Table A4-1 shows the descriptive statistics of variables at the video level analysis. It should be noted that, different from creator-week level analysis, there are 20,766 observations at the video level.

Table A4-1

Descriptive statistics at the video level analysis.

Variable	Observation	Mean	Standard deviation	Min	Max
LoudnessVariability	20,766	4045.195	2608.065	2.602	20,931.99
VocalPitch	20,766	280.352	163.117	76.152	2205
Length	20,766	702.425	835.687	2.05	18,960.23
ContentQuality	20,766	3.263	5.921	−50	100
VideoCount	20,766	1.924	1.823	1	33
View	20,766	94,571.67	272,475.9	3	6,589,119
Speed	20,766	258.445	191.36	0.010	1735.08
Zcr	20,766	1028.797	10,809.29	486	506,953
Like	20,766	6854.089	22,964.54	0	719,783
FollowerChangeR	20,766	0.005	0.026	−0.924	1.924

To examine the effect of acoustic characteristics on the number of likes at the video level, the following model is adopted:

$$\text{Ln}(\text{Like}_j + 1) = \text{Ln}(\text{LV}_j + 1) + \text{Ln}(\text{PI}_j + 1) + \text{VoiceControls}_j + \text{Ln}(\text{View}_j + 1) + \text{ContentQuality}_j + \sum (\text{Category Dummies}_j) + \sum (\text{CreatorDummies}_j) + \alpha_j + \epsilon_j$$

To examine the effect of acoustic characteristics on the changes in the number of followers at the video level, the following model is adopted:

$$\text{Ln}(\text{FollowerChangeR}_j + 1) = \text{Ln}(\text{LV}_j + 1) + \text{Ln}(\text{PI}_j + 1) + \text{VoiceControls}_j + \text{Ln}(\text{View}_j + 1) + \text{ContentQuality}_j + \sum (\text{Category Dummies}_j) + \sum (\text{CreatorDummies}_j) + \alpha_j + \epsilon_j$$

Ln(Like_j + 1) is the log-transformed number of likes video *j* receives. Ln(FollowerChangeR_j + 1) is the log-transformed incremental ratio of followers for video *j*. We obtained FollowerChangeR_j by calculating the incremental ratio of followers video *j* receives within seven days after it is published. Ln(LV_j + 1) and Ln(PI_j + 1) are the log-transformed loudness variability and the log-transformed vocal pitch for video *j*. The results are reported in Table A4-2, which are no much difference with the main findings.

Table A4-2

Results of video level analysis.

	Column 1	Column 2	Column 3	Column 4
	DV = Ln (Like _j + 1)		DV = Ln(FollowerChangeR _j + 1)	
Ln(LV _j + 1)	0.291*** (14.07)	0.392*** (18.68)	0.0004** (2.34)	0.0004** (2.31)
Ln(PI _j + 1)	0.595*** (14.97)	0.244*** (15.39)	−0.0001** (−2.07)	−0.0001* (−2.07)
Ln(Length _j + 1)	0.256*** (16.99)	0.228*** (12.83)	−0.0004*** (−3.31)	−0.0004*** (−3.13)
Ln(VideoCount _j + 1)	−0.127*** (−6.80)	−0.125*** (−6.66)	0.0001*** (6.04)	0.0001*** (6.21)
Ln(Length _j + 1)*Ln(LV _j + 1)		0.235*** (13.39)		
Ln(Length _j + 1)*Ln(PI _j + 1)		−0.091*** (−2.61)		

(continued on next page)

Table A4-2 (continued)

	Column 1	Column 2	Column 3	Column 4
	DV = $\ln(\text{Like}_j + 1)$		DV = $\ln(\text{FollowerChangeR}_j + 1)$	
$\ln(\text{VideoCount}_j + 1) * \ln(\text{LV}_j + 1)$				0.0007* (1.86)
$\ln(\text{VideoCount}_j + 1) * \ln(\text{PI}_j + 1)$				0.0001* (1.85)
Control variables	Included	Included	Included	Included
Creator dummies	YES	YES	YES	YES
Category dummies	YES	YES	YES	YES
Constant	−20.027*** (−13.08)	−5.547*** (−2.76)	0.001** (2.17)	0.001* (1.95)
Adjusted R-squared	0.174	0.186	0.501	0.502
Observation	20,766	20,766	20,766	20,766

Appendix A5: Alternative measurement intervals of dependent variables

Considering it is a cumulative process to be familiar with and get used to ones' voice, it may take time for video creators' acoustics characteristics to exert effects and to be converted into the number of likes and the increase in the number of followers (more like a process of quantitative alteration to qualitative change). Thus, a robustness check with alternative time intervals (with the main independent variables remained) was conducted to verify the results. Specifically, the dependent variables were one-week lagged and re-analyzed as shown in Table A5. The results are consistent with the main results, suggesting that our findings are not sensitive to the time intervals in the dependent variables. However, when the time intervals are changed, the impacts of loudness variability and vocal pitch on the number of likes are not as significant as in the main results. It suggests that videos released weeks ago may not affect the number of likes a creator receives through the effective use of voices. This finding also coincides with our previous argument that the number of likes reflects the short-term engagement of viewers.

Table A5

Results with one-week lagged dependent variables.

	DV = $\ln(\text{Like}_{i,t+1} + 1)$		DV = $\ln(\text{FollowerChangeR}_{i,t+1} + 1)$	
	Column 1	Column 2	Column 1	Column 2
$\ln(\text{LV}_{i,t} + 1)$	0.028 (0.95)	0.025 (0.74)	0.0001* (1.82)	0.0001* (1.77)
$\ln(\text{PI}_{i,t} + 1)$	0.037 (1.22)	0.047 (1.51)	−0.0007** (−2.13)	−0.0007** (−2.08)
$\ln(\text{VideoCount}_{i,t} + 1)$	−0.082*** (−5.27)	−0.083*** (−7.57)	0.005*** (3.33)	0.002** (1.99)
$\ln(\text{Length}_{i,t} + 1)$	0.089** (2.28)	0.081** (2.42)	−0.0005 (−1.17)	−0.0006 (−1.23)
$\ln(\text{Length}_{i,t} + 1) * \ln(\text{LV}_{i,t} + 1)$		0.007 (1.22)		
$\ln(\text{Length}_{i,t} + 1) * \ln(\text{PI}_{i,t} + 1)$		−0.035 (−1.04)		
$\ln(\text{VideoCount}_{i,t} + 1) * \ln(\text{LV}_{i,t} + 1)$				0.0004* (1.73)
$\ln(\text{VideoCount}_{i,t} + 1) * \ln(\text{PI}_{i,t} + 1)$				0.0001* (1.79)
Control variables	Included	Included	Included	Included
Category dummies	YES	YES	YES	YES
Constant	−0.671*** (−7.96)	0.644*** (−6.69)	0.002*** (3.84)	0.003*** (4.86)
Adjusted R-squared	0.074	0.074	0.144	0.145
Observation	18,307	18,307	18,307	18,307
Week-fixed effect	YES	YES	YES	YES
Individual-fixed effect	YES	YES	YES	YES

Note: Robust T-statistics in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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